

GIT LOCAL TO GITHUB CONNECTION USING SSH

STEP 1: CREATE LOCAL REPOSITORY

- Create a new folder: `mkdir git-course`
- Go inside folder: `cd git-course`
- Initialize git: `git init`
- Create file: `echo "git-course-demo" >> README.md`
- Add file: `git add README.md`
- Commit file: `git commit -m "first commit"`
- Rename branch to main: `git branch -M main`

STEP 2: GENERATE SSH KEY

- Run command: `ssh-keygen -o`
- This creates two files inside `C:\Users\YourName\.ssh\`
- Private Key: `id_ed25519`
- Public Key: `id_ed25519.pub`
- View public key: `type id_ed25519.pub`
- Copy the entire key

STEP 3: ADD SSH KEY TO GITHUB

- Open: <https://github.com/settings/keys>
- Click: New SSH Key
- Paste copied public key
- Click Save

STEP 4: ADD REMOTE REPOSITORY

- Copy SSH URL from GitHub (Example: `git@github.com:username/git-course.git`)
- Run: `git remote add origin git@github.com:username/git-course.git`

STEP 5: PUSH CODE TO GITHUB

- Run: `git push -u origin main`

- If prompted, type: yes
- Your local repository is now connected to GitHub using SSH

SSH vs HTTPS (Easy Comparison)

- SSH: Uses SSH key authentication
- SSH: No need to enter password every time
- SSH: One-time setup, more secure for developers
- HTTPS: Uses username + Personal Access Token
- HTTPS: May ask credentials during push
- HTTPS: Easier for beginners (no SSH setup required)

IMPORTANT GIT COMMANDS SUMMARY

- `git init`
- `git add .`
- `git commit -m "message"`
- `git branch -M main`
- `git remote add origin`
- `git push -u origin main`